GC University Lahore

Department of Computer Science

Course: Software Engineering (CS-2105 / CS-2111)

Paper 1: Software Engineering Past Paper (Subjective)

Time Allowed: 150 minutes

Max. Marks: 40

Note: Attempt any Four Questions.

Question No. 2 [2 + 2 + 2 + 2 + 2 = 10 Marks]

Differentiate between the following terms with clear, concise definitions:

1. **Extend** and **include** relationships in use cases.
2. **Composition** and **aggregation** in class associations.
3. **Validation** and **verification** in software quality assurance.
4. **Inception** and **elicitation** in requirement engineering.
5. **Black box** and **white box** testing strategies.

Question No. 3 [5 + 5 = 10 Marks]

- **a.** Elaborate on software framework activities with a suitable real-world example.
- **b.** Elaborate on the four phases of Extreme Programming (XP).

Question No. 4 [5 + 5 = 10 Marks]

- **a.** Distinguish at least **TWO** functional and **TWO** non-functional requirements of an online shopping website that allows customers to browse, search, and buy various products. It works on the traditional system of filling a cart and making payments online with shipment facilities. The online website requires 24/7 availability, and a response to any user query must be made within 2 hours.
- **b.** Design a complete **Use Case Diagram** for the system described above.

Question No. 5 [5 + 5 = 10 Marks]

- **a.** Explain the seven steps of the Requirement Engineering process.
- **b.** Describe the differences and relationship between **Unit Testing** and **Integration Testing**.

CASE STUDY: Retail Store Management System

The Retail Store Management System is designed for managing store processes: ordering, arranging, and selling goods. The Retailer checks for the availability of goods in the store. If the stock of goods is less than a minimum threshold, the retailer places an order for goods. While ordering goods, the goods are received at the store. The retailer then arranges them by product type or by price, and makes the payment. If the stock of goods is already available, the retailer arranges them directly for selling.

The retailer then sells the goods directly to the customer. The customer buys the items from the retailer. The retailer prepares a bill for the goods purchased by the customer, and receives the amount either by credit or by cash. The supplier supplies the goods to the store within the system.
The overall system is used to manage inventory in the store.

Considering the above scenario, design:

- **a.** A detailed **Class Diagram**.
- **b.** A functional **Activity Diagram**.

Paper 2: Final Term Examination Fall 2021

Course Code: CS-2105

Semester: V

Total Marks: 40

Time Allowed: 150 minutes

NOTE: Attempt any 4 Questions. Each question carries equal marks (10 Marks each).

Question No. 2 [6 + 4 = 10 Marks]

- **a)** Write a detailed note on the Core Agility Principles.
- **b)** What is the importance of Modeling within the Software Process Framework?

Question No. 3 [6 + 4 = 10 Marks]

- **a)** Which one is more critical to software success: **Validation** or **Verification**? Define both terms with appropriate examples.
- **b)** Is it possible to combine different process models (hybrid models)? If so, provide an illustrative example.

Question No. 4 [10 Marks]

Read the project description carefully and map it to the **Spiral Model** process. Define the specific phases involved within this process model framework.

> includes registration of patients and storing their disease details in the system. It will also contain doctors' information and will digitalize the whole billing system. The software has the facility to give a unique ID to every patient and stores the details of every patient and staff member automatically. It includes a search facility to know the current status of each room. Users can search for the availability of a doctor and the details of a patient using their unique ID.

The Hospital Management System can be accessed by entering a respective username and password. It is accessible either by an administrator or a receptionist. Only authorized personnel can add data to the database. The data can be retrieved easily. The interface is highly user-friendly. The data are well-protected, and data processing is fast, accurate, and relevant.

Question No. 5 [8 + 2 = 10 Marks]

- **a)** What is meant by **Umbrella Activities**? Give at least four examples.
- **b)** Describe what an **analysis pattern** is in your own words.

Question No. 6 [10 Marks]

You have been asked to build **one** of the following systems:

- **a)** A network-based course registration system for your university.
- **b)** A Web-based order-processing system for a computer store.
- **c)** A simple invoicing system for a small business.
- **d)** An Internet-based cookbook that is built into an electric range or microwave.

Select only one system from above and design its:

1. Use-Case Diagram
2. Activity Flow Diagram
3. Sequence Diagram

Paper 3: Final Examination 2023

Semester: 5^{th}

Course Code: CS-2105

Total Marks: 40

Time Allowed: 155 minutes

NOTE: Attempt any Four Questions. All questions carry equal marks.

Question No. 2 [5 + 5 = 10 Marks]

- **(a)** Name the phases of Extreme Programming (XP). Explain the core actions associated with each activity.

  performed during UP phases.

Question No. 3 [7 + 3 = 10 Marks]

- **(a)** Briefly discuss the seven tasks of Requirement Engineering.
- **(b)** Differentiate between **cohesion** and **coupling**. Why is "High Cohesion, Low Coupling" preferred?

Question No. 4 [5 + 5 = 10 Marks]

- **(a)** Differentiate between the following:
  1. White box and Black box testing.
  2. Alpha and Beta testing.
- **(b)** Briefly describe three different techniques of **Black box testing**.

Question No. 5 [10 Marks]

Create a comprehensive **Use Case Diagram** based on the following functional requirements:

> **Online University Registration System:** > The system should enable the staff of each academic department to examine the courses offered by their department, add and remove courses, and change course metadata (e.g., the maximum number of students permitted).
>
> It must permit students to examine currently available courses, add and drop courses to and from their schedules, and inspect the courses for which they are already enrolled. Departmental staff should be able to print various reports regarding courses and the students enrolled in them. The system must automatically ensure that no student takes too many courses and that students who have any unpaid fees are blocked from registering.

Question No. 6 [8 + 2 = 10 Marks]

- **(a)** Briefly describe each of the four fundamental elements of software **design models**.
- **(b)** What is **refactoring**? Explain its benefits in code maintenance.

Paper 4: Final Examination 2025 (Evening)

Semester: V

Course Code: CS-2105

Max. Marks: 40

Time Allowed: 160 minutes

Note: Attempt any 4 questions. All Questions carry equal Marks. This paper is closedbook.

Below are five software applications that are to be designed and developed by a software house. As a software engineer, choose and recommend the most suitable software process model for each. Justify your choice for each case:

1. A well-understood, standard data processing application.
2. An online system of order transactions with rapidly changing and conflicting requirements.
3. A full-scale operating system.
4. A test-driven Dropbox desktop client application (strictly fixed development time of 60–90 days).
5. A highly critical Nuclear Reactor Control System.

Q. 3 [10 Marks]

Compute the **Function Point (FP)**, **Productivity (FP/pm)**, **Documentation (Pages/FP)**, and **Cost per Function ($/FP)** given the following collection of project metrics:

1. Number of user inputs (Average) = 24
2. Number of user outputs (Low) = 46
3. Number of inquiries (Complex) = 8
4. Number of files (Average) = 4
5. Number of external interfaces (Low) = 2
6. Effort = 36.9 person-months (pm)
7. Technical documents = 265 pages
8. User documents = 122 pages
9. Labor cost = $7,744/month

Processing Complexity Factors (F_i): The fourteen value adjustment factors are: 4,1,0,3,3,5,4,4,3,3,2,2,4,5.

(Hint: Use standard function point calculations where $FP = UAF \times [0.65 + 0.01 \times \sum F_i]$).

Q. 4 [5 + 5 = 10 Marks]

- **(a)** Describe the key technical differences between **black box testing** and **white box testing**.
- **(b)** Mr. Ahmad is a software engineer who insists on writing test cases before writing any code. He believes this approach reveals specification problems at an early stage and permits parallel development. Moreover, he believes that if the code already exists, it biases the tester into the same logical paths, causing them to make the same errors as the programmer.

  Your Task: Analyze whether this test-driven approach is more applicable to black-box tests, white-box tests, or equally applicable to both. **State your position clearly** before providing your justifications.

Q. 5 [5 + 5 = 10 Marks]

- **(b)** Explain the concept of **Dependency Injection (DI)** and describe its benefits in creating loosely coupled systems.